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(54) **WORKPIECE PROCESSING APPARATUS
WITH THERMAL PROCESSING SYSTEMS**

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(57) **ABSTRACT**

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A processing apparatus for a thermal treatment of a work-
piece is presented. The processing apparatus includes a
processing chamber, a workpiece support disposed within
the processing chamber, a gas delivery system configured to
flow one or more process gases into the processing chamber
from the a first side of the processing chamber, one or more
radiative heating sources disposed on the second side of the
processing chamber, one or more dielectric windows dis-
posed between the workpiece support and the one or more
radiative heating sources, a rotation system configured to
rotate the one or more radiative heating sources, and a
workpiece temperature measurement system configured at a
temperature measurement wavelength range to obtain a
measurement indicative of a temperature of a back side of
the workpiece.

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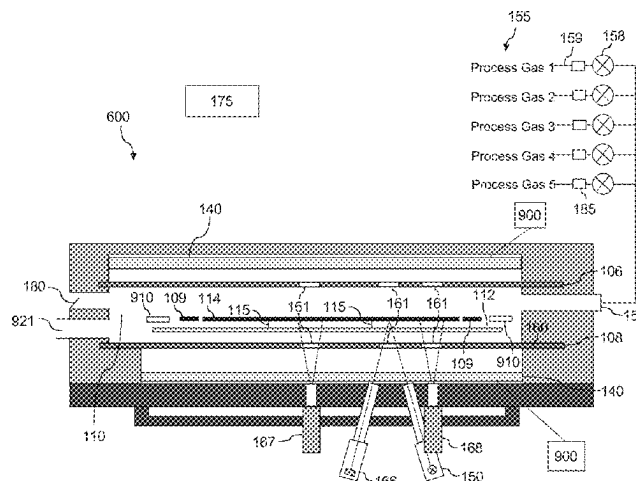
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See application file for complete search history.

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H05B 1/02 (2006.01)
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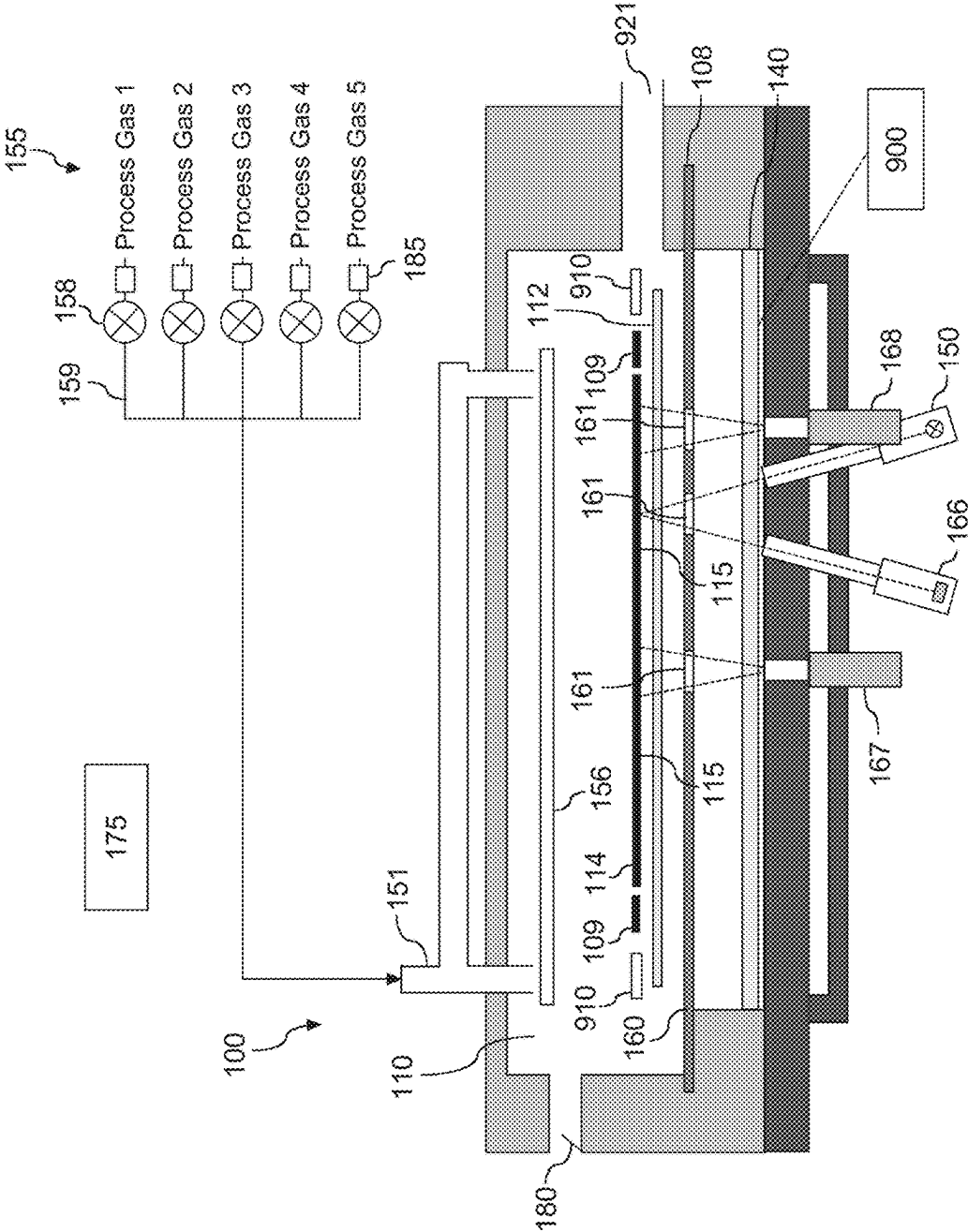
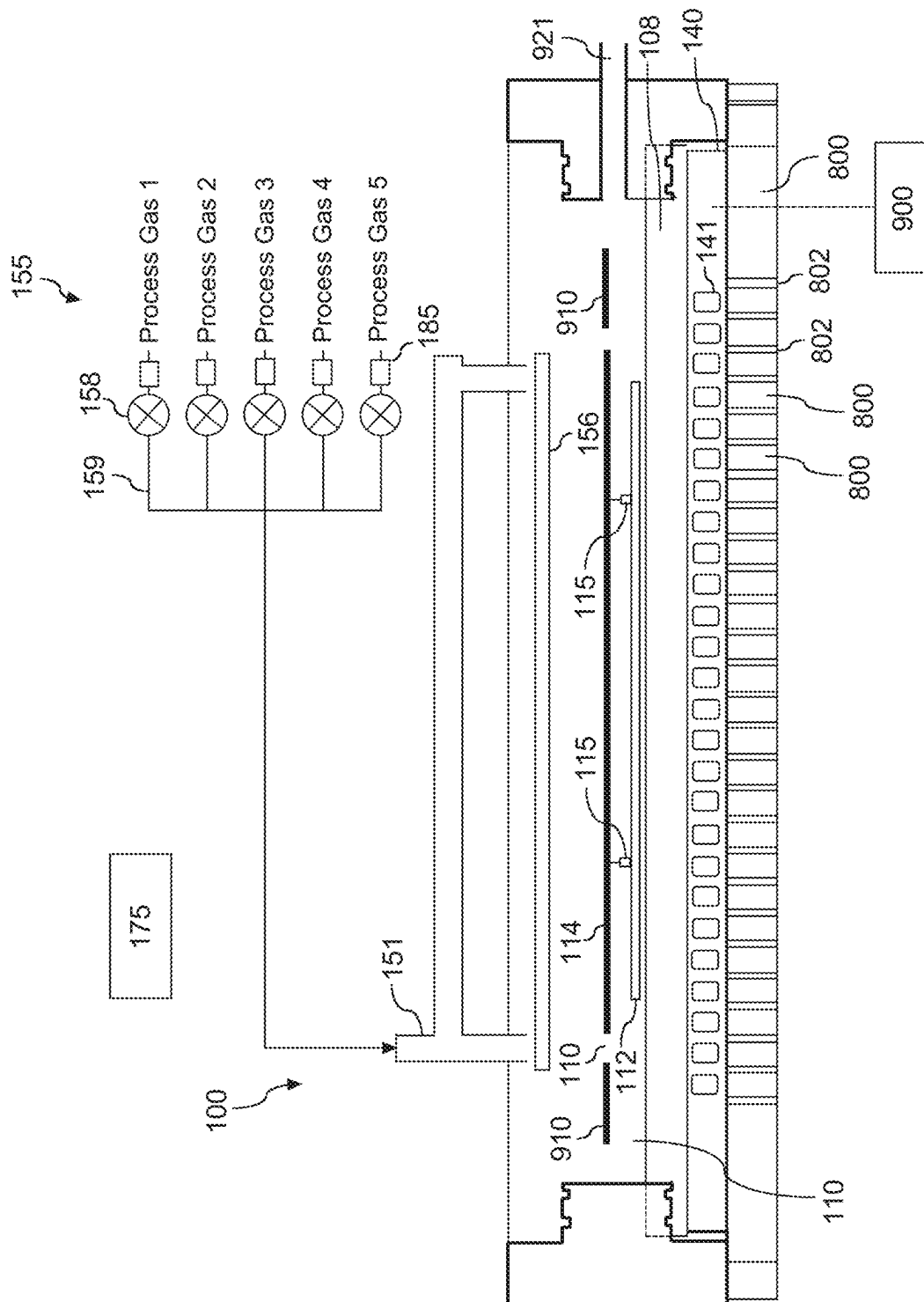
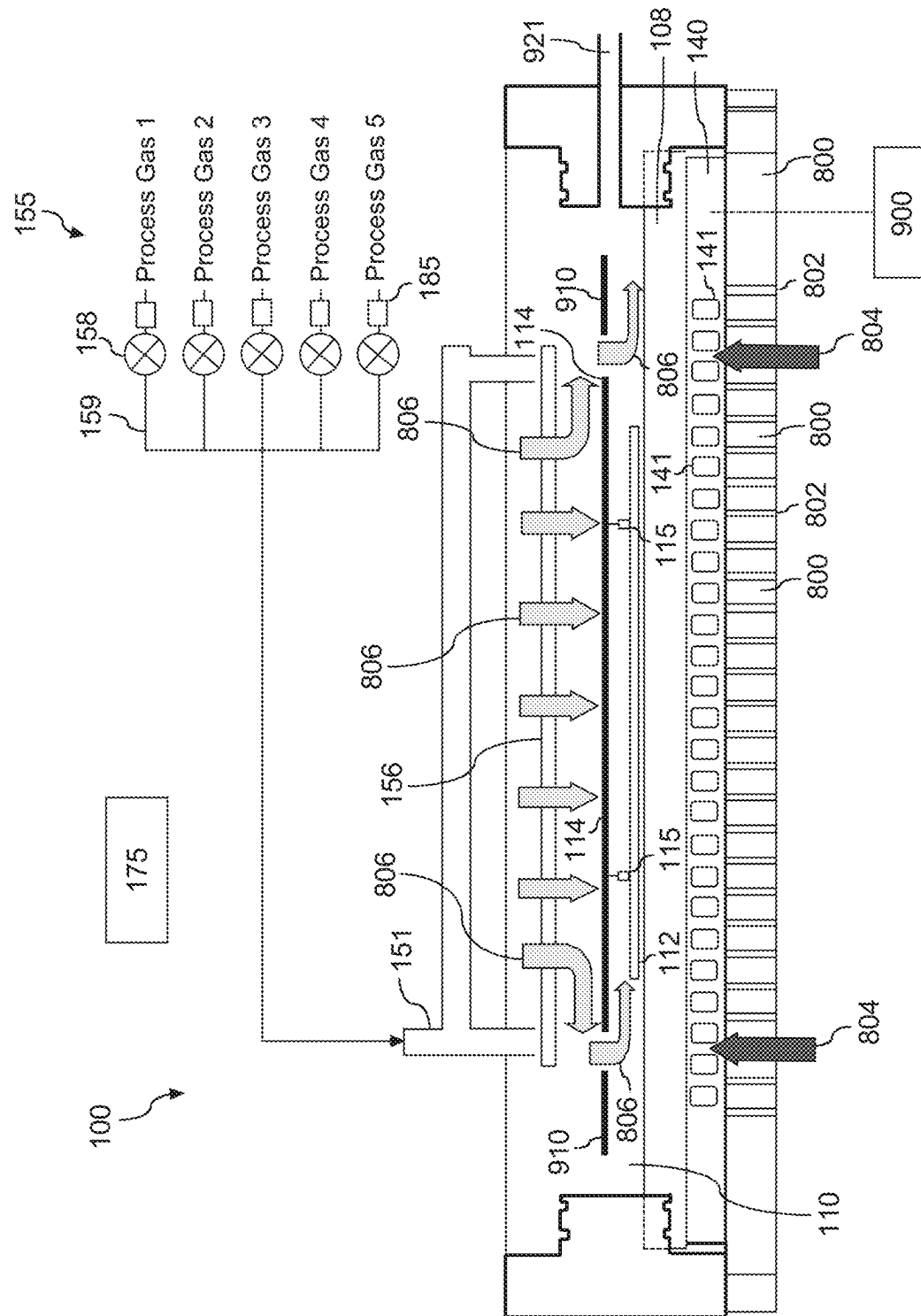
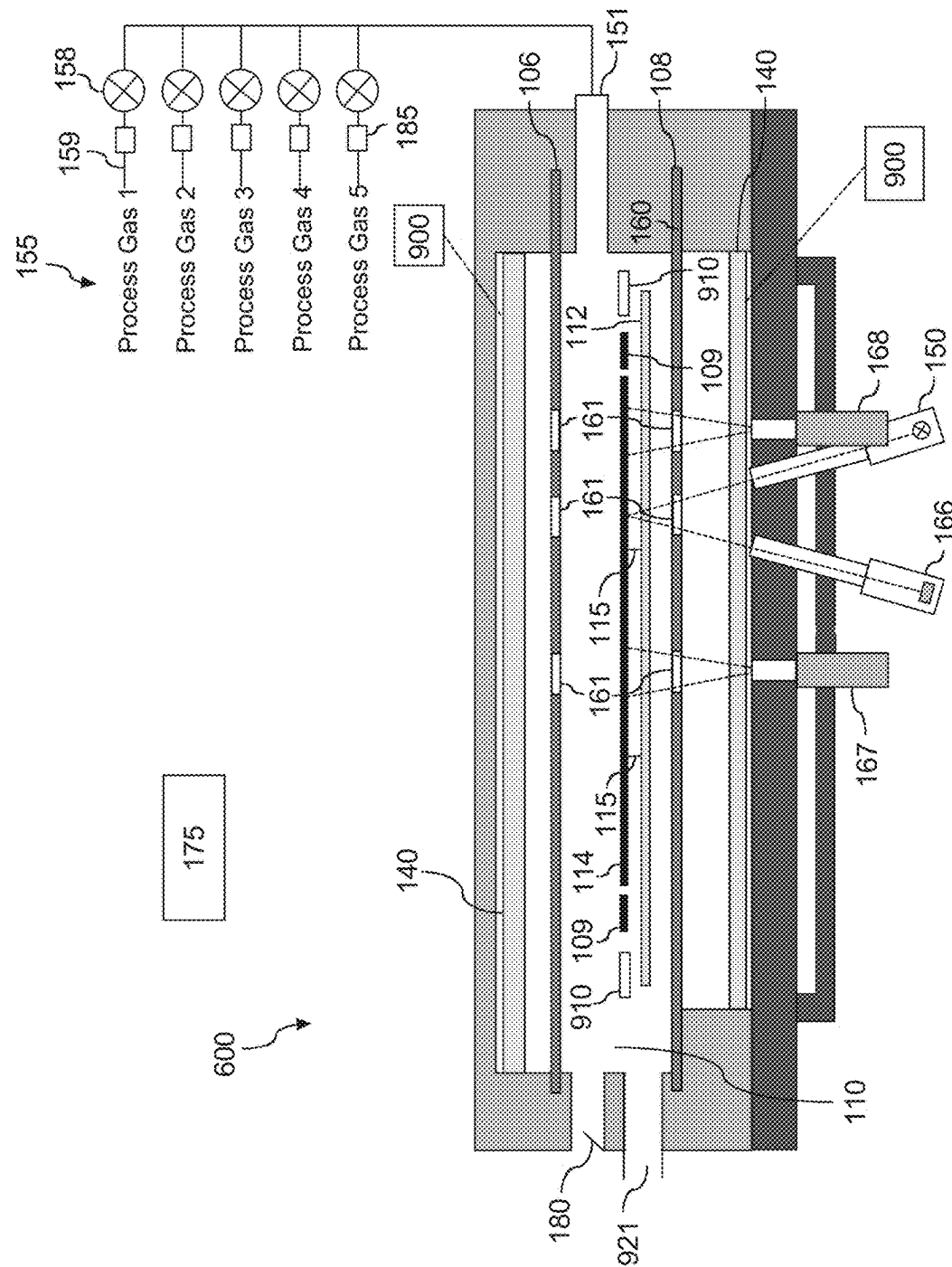


FIG. 1

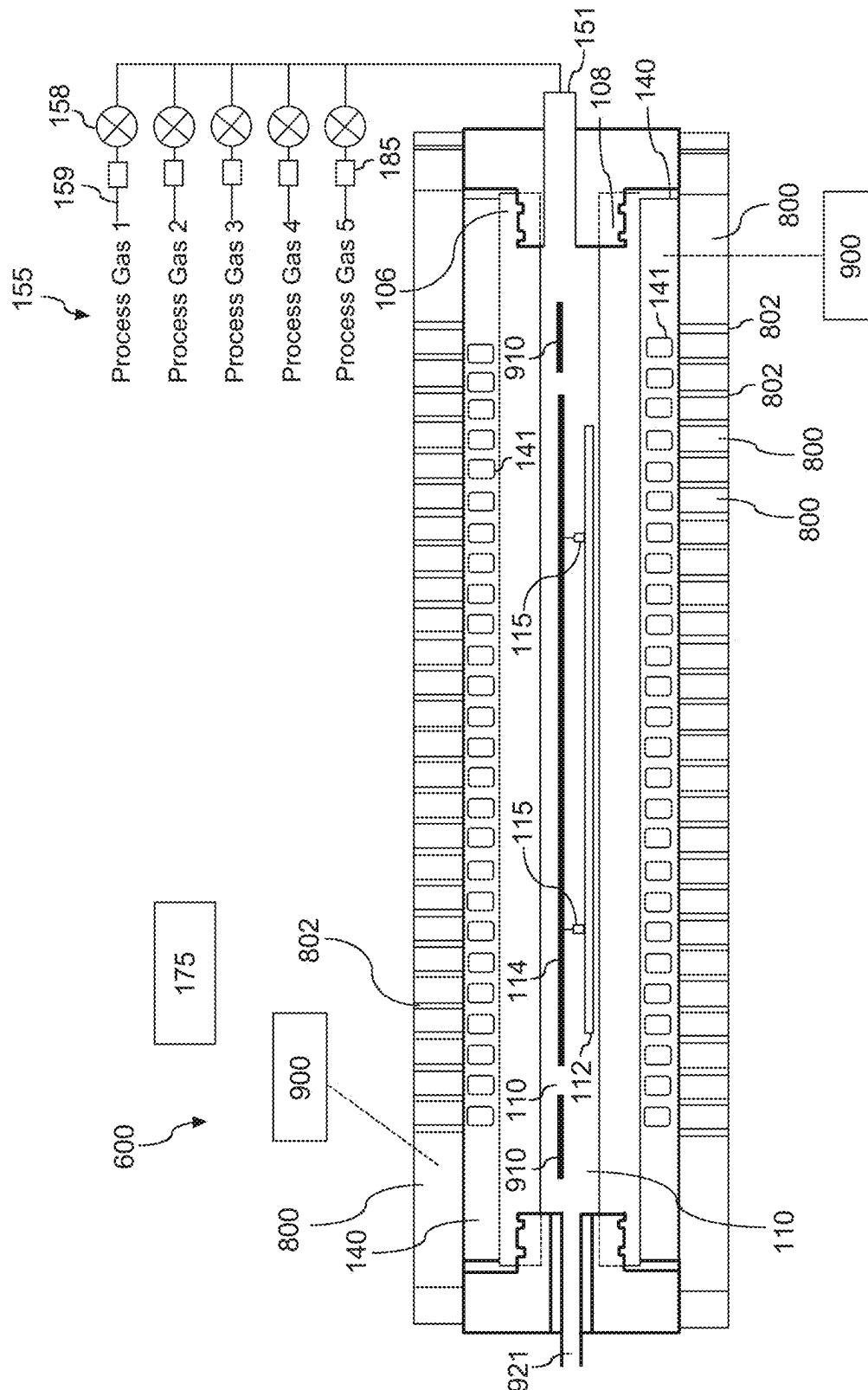
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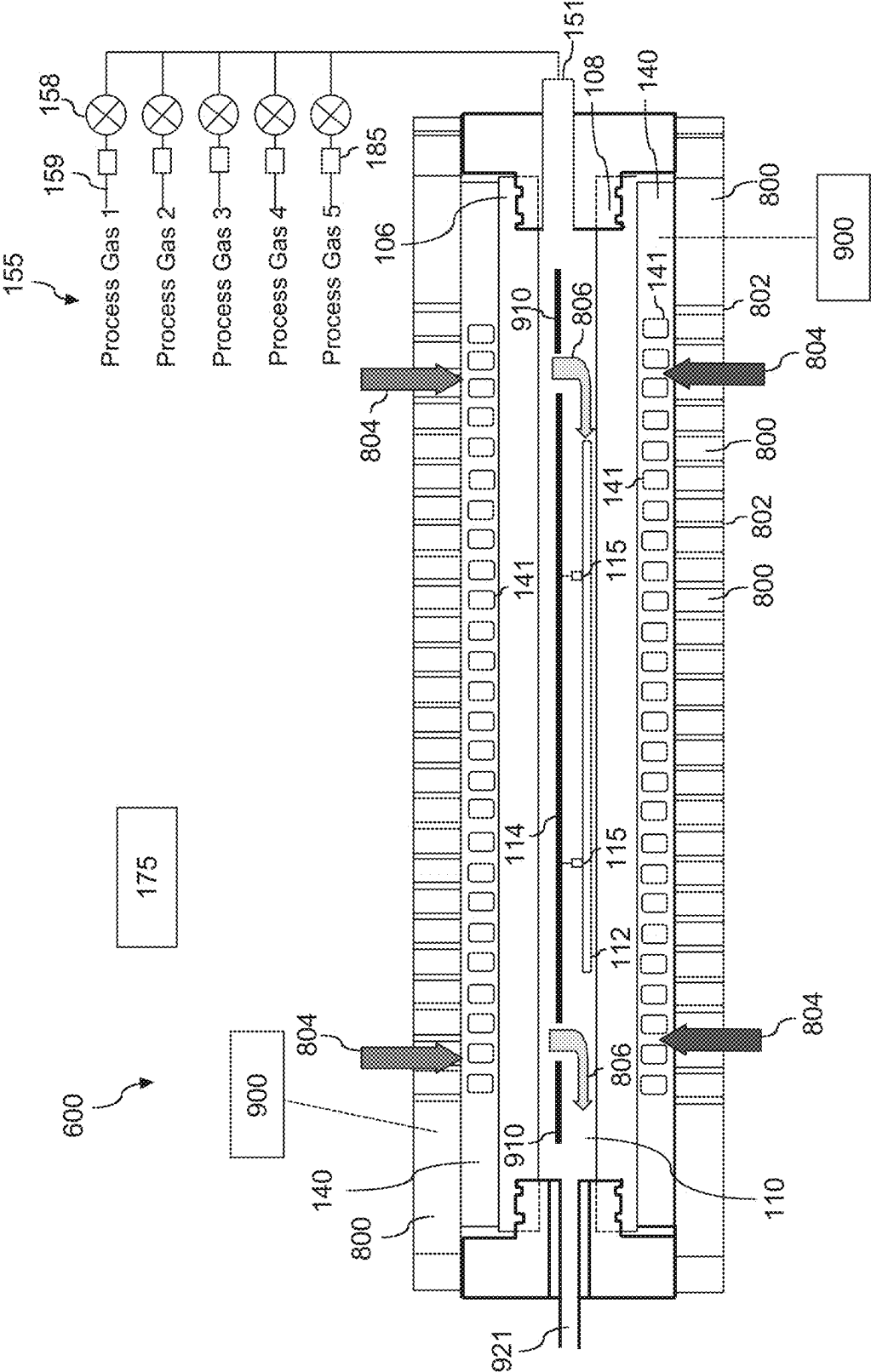


FIG. 6

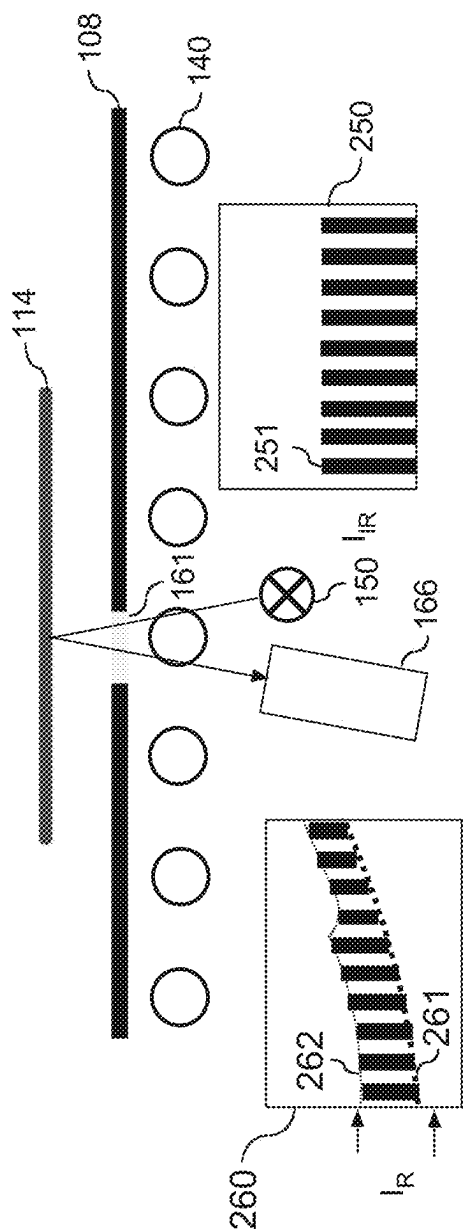


FIG. 7

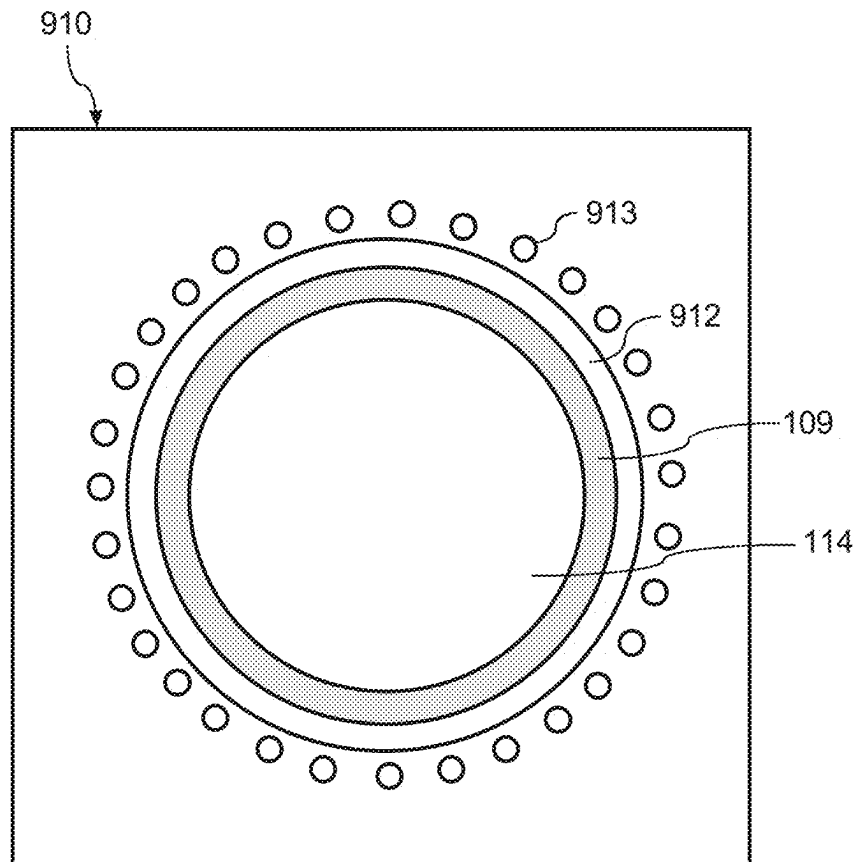


FIG. 8

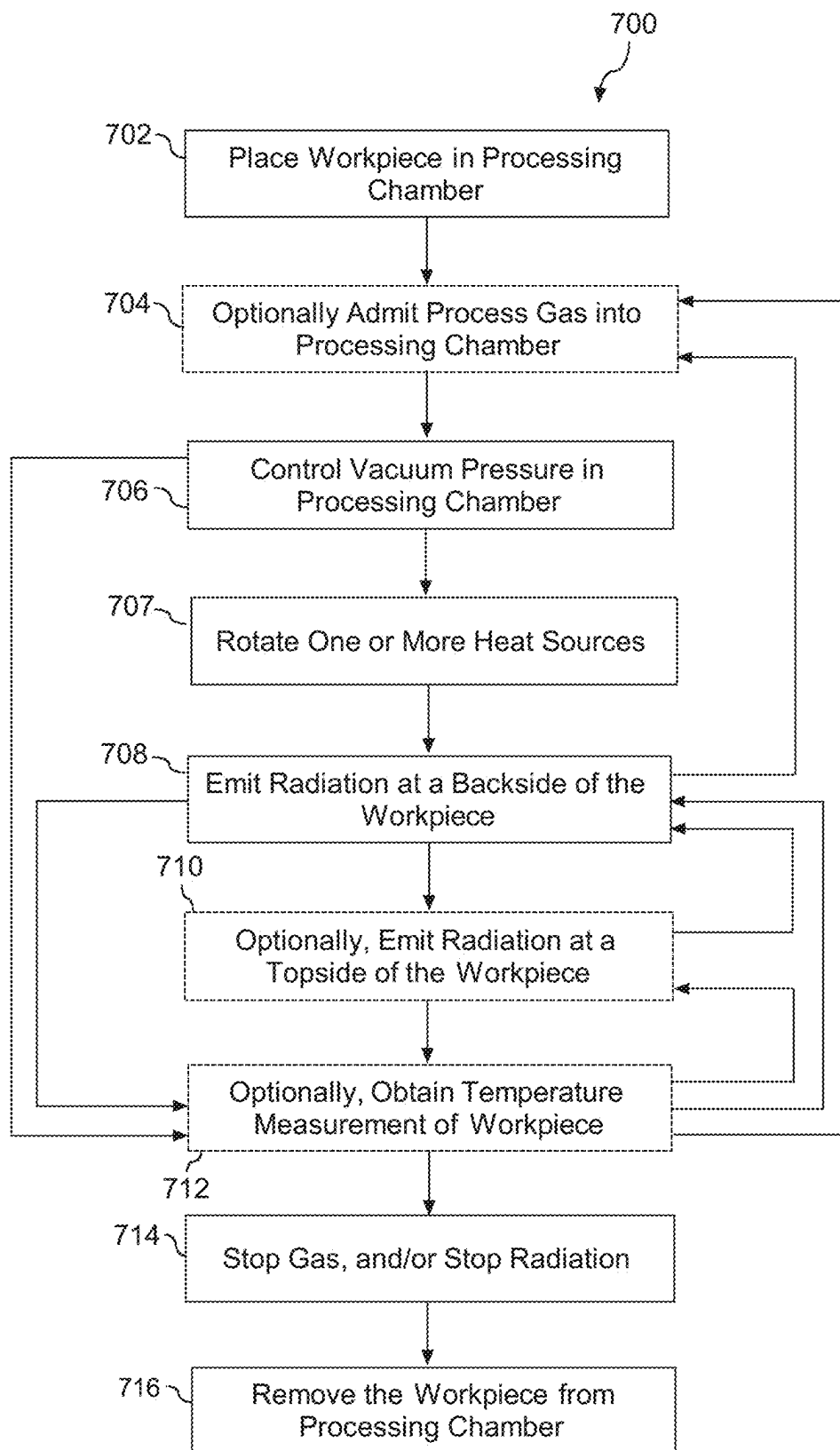


FIG. 9

WORKPIECE PROCESSING APPARATUS WITH THERMAL PROCESSING SYSTEMS

PRIORITY CLAIM

The present application is a divisional of U.S. Non-Provisional application Ser. No. 17/550,154 filed on Dec. 14, 2021, which claims the benefit of priority of U.S. Provisional Application Ser. No. 63/130,969, titled "Workpiece Processing Apparatus with Thermal Processing Systems," filed on Dec. 28, 2020. Applicant claims priority to and the benefit of each of such applications and incorporates all such applications herein by reference in its entirety.

FIELD

The present disclosure relates generally to semiconductor processing equipment, such as equipment operable to perform thermal processing of a workpiece.

BACKGROUND

Thermal processing is commonly used in the semiconductor industry for a variety of applications, including and not limited to post-implant dopant activation, conductive and dielectric materials anneal, in addition to materials surface treatments including oxidation and nitridation. Generally, a thermal processing chamber as used herein refers to a device that heats workpieces, such as semiconductor workpieces. Such devices can include a support plate for supporting one or more workpieces and an energy source for heating the workpieces, such as heating lamps, lasers, or other heat sources. During heat treatment, the workpiece(s) can be heated under controlled conditions according to a processing regime. Many thermal treatment processes require a workpiece to be heated over a range of temperatures so that various chemical and physical transformations can take place as the workpiece is fabricated into a device(s). During rapid thermal processing, for instance, workpieces can be heated by an array of lamps to temperatures from about 300° C. to about 1,200° C. over time durations that are typically less than a few minutes. Improvement in thermal processing devices are desirable to effectively measure and control workpiece temperature with a variety of desired heating schemes.

SUMMARY

Aspects and advantages of embodiments of the present disclosure will be set forth in part in the following description, or may be learned from the description, or may be learned through practice of the embodiments.

Example aspects of the present disclosure are directed to a processing apparatus for processing a workpiece. The processing apparatus includes a processing chamber, having a first side and a second side opposite from the first side of the processing chamber; a workpiece support disposed within the processing chamber, the workpiece support configured to support the workpiece, wherein the back side of the workpiece faces the workpiece support; a gas delivery system configured to flow one or more process gases into the processing chamber from the first side of the processing chamber; one or more radiative heating sources disposed on the second side of the processing chamber, the one or more radiative heating sources configured to heat the workpiece from the back side of the workpiece; a rotation system configured to rotate the one or more radiative heating

sources; one or more dielectric windows disposed between the workpiece support and the one or more radiative heating sources; and a workpiece temperature measurement system configured at a temperature measurement wavelength range to obtain a temperature measurement indicative of a temperature of the back side of the workpiece.

Example aspects of the present disclosure are also directed to a processing apparatus for processing a workpiece. The processing apparatus includes a processing chamber having a first side and a second side opposite from the first side of the processing chamber; a workpiece support disposed within the processing chamber, the workpiece support configured to support the workpiece, wherein the back side of the workpiece faces the workpiece support; a gas delivery system configured to flow one or more process gases into the processing chamber; a first group of one or more radiative heating sources disposed on the first side of the processing chamber, the first group of one or more radiative heating sources configured to heat the workpiece from the top side of the workpiece; a second group of one or more radiative heating sources disposed on the second side of the processing chamber, the second group of one or more radiative heating sources configured to heat the workpiece from the back side of the workpiece; a first dielectric window disposed between the first group of one or more radiative heating sources disposed on the first side of the processing chamber and the workpiece; a second dielectric window disposed between the second group of one or more radiative heating sources disposed on the second side of the processing chamber and the workpiece support; and a rotation system configured to rotate the first group of one or more radiative heating sources and/or the second group of one or more radiative heating sources; and a workpiece temperature measurement system configured at a temperature measurement wavelength range to obtain a temperature measurement indicative of a temperature of the back side of the workpiece.

These and other features, aspects and advantages of various embodiments will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present disclosure and, together with the description, serve to explain the related principles.

BRIEF DESCRIPTION OF THE DRAWINGS

Detailed discussion of embodiments directed to one of ordinary skill in the art are set forth in the specification, which makes reference to the appended figures, in which:

FIG. 1 depicts an example processing system according to example aspects of the present disclosure;

FIG. 2 depicts an example processing system according to example aspects of the present disclosure;

FIG. 3 depicts an example processing system according to example aspects of the present disclosure;

FIG. 4 depicts an example processing system according to example aspects of the present disclosure;

FIG. 5 depicts an example processing system according to example aspects of the present disclosure;

FIG. 6 depicts an example processing system according to example aspects of the present disclosure;

FIG. 7 depicts an example temperature measurement system according to example embodiments of the present disclosure;

FIG. 8 depicts an example pumping plate according to example aspects of the present disclosure; and

FIG. 9 depicts an example flowchart of a method according to example aspects of the present disclosure.

DETAILED DESCRIPTION

Reference now will be made in detail to embodiments, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the embodiments, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made to the embodiments without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that aspects of the present disclosure cover such modifications and variations.

Various workpiece processing treatments can require heat treatment under controlled conditions in a processing chamber. For instance, thermal treatment of the workpiece can include heating either the back side or the top side of a workpiece with one or more radiative heat sources. During heating, it is often desirable to rotate the workpiece such that the workpiece experiences heating uniformity. Accordingly, devices and mechanisms for rotating the workpiece itself have been developed. However, improved devices for uniformly heating the workpiece are still needed.

Accordingly, aspects of the present disclosure provide a number of technical effects and benefits. For instance, the processing apparatus provided herein allows for the ability to rotate the radiative heat sources. Accordingly, the workpiece does not have to be rotated in the thermal chamber during processing. Rotation of the radiative heat sources themselves allows for uniform heat distribution during processing without having to rotate the workpiece. For example, certain devices used to rotate the workpiece can cause damage to the workpiece. Thus, the presently disclosed system can reduce workpiece damage during processing. Further, the processing apparatus is capable of obtaining an accurate temperature measurement of the workpiece during processing.

Variations and modifications can be made to these example embodiments of the present disclosure. As used in the specification, the singular forms “a,” “and,” and “the” include plural referents unless the context clearly dictates otherwise. The use of “first,” “second,” “third,” etc., are used as identifiers and are not necessarily indicative of any ordering, implied or otherwise. Example aspects may be discussed with reference to a “substrate,” “workpiece,” or “workpiece” for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that example aspects of the present disclosure can be used with any suitable workpiece. The use of the term “about” in conjunction with a numerical value refers to within 20% of the stated numerical value.

Example embodiments of a processing apparatus will now be discussed with reference to FIGS. 1-6. As shown in FIG. 1, according to example aspects of the present disclosure, the apparatus 100 can include a gas delivery system 155 configured to deliver process gas to the processing chamber 110, for instance, via a gas distribution channel 151. The gas delivery system can include a plurality of feed gas lines 159. The feed gas lines 159 can be controlled using valves 158 and/or gas flow controllers 185 to deliver a desired amount of gases into the processing chamber as process gas. The gas delivery system 155 can be used for the delivery of any suitable process gas. Example process gases include, oxy-

gen-containing gases (e.g., O₂, O₃, N₂O, H₂O), hydrogen-containing gases (e.g., H₂, D₂), nitrogen-containing gas (e.g., N₂, NH₃, N₂O), fluorine-containing gases (e.g., CF₄, C₂F₄, CHF₃, CH₂F₂, CH₃F, SF₆, NF₃), hydrocarbon-containing gases (e.g., CH₄), or combinations thereof. Other feed gas lines containing other gases can be added as needed. In some embodiments, the process gas can be mixed with an inert gas that can be called a “carrier” gas, such as He, Ar, Ne, Xe, or N₂. A control valve 158 can be used to control a flow rate of each feed gas line to flow a process gas into the processing chamber 110. In embodiments, the gas delivery system 155 can be controlled with gas flow controllers 185.

As shown in FIG. 1, the gas delivery system can be disposed about a first side of the processing chamber 110, such as the top side of the processing chamber. Accordingly, the gas delivery system 155 can provide process gases to the top side of the processing chamber. Accordingly, process gas delivered by the gas delivery system 155 is first exposed to the top side of the workpiece 114 in the processing chamber 110.

In embodiments, the processing apparatus 100 can include one or more gas distribution plates 156 disposed about the first side of the processing chamber 110. The one or more gas distribution plates 156 can be used to more uniformly disperse process gases in the processing chamber 110. Process gases can be delivered by the distribution channel 151 and pass through one or more gas distribution plates 156 to more uniformly and evenly distribute gas in the processing chamber 110, thus ensuring that the top side of the workpiece 114 is uniformly exposed to process gases. In embodiments, the gas distribution plates can include a plurality of apertures or channels configured to facilitate uniform distribution of process gases in the processing chamber 110.

The workpiece 114 to be processed is supported in the processing chamber 110 by the workpiece support 112. Workpiece 114 can be or include any suitable workpiece, such as a semiconductor workpiece, such as a silicon wafer. In some embodiments, workpiece 114 can be or include a doped silicon wafer. For example, a silicon wafer can be doped such that a resistivity of the silicon wafer is greater than about 0.1 Ω·cm, such as greater than about 1 Ω·cm.

Workpiece support 112 can be or include any suitable support structure configured to support workpiece 114 in processing chamber 110. For example, the workpiece support 112 can be a workpiece support 112 operable to support a workpiece 114 during thermal processing (e.g., a workpiece support plate). In some embodiments, workpiece support 112 can be configured to support a plurality of workpieces 114 for simultaneous thermal processing by a thermal processing system. In some embodiments, workpiece support 112 can rotate workpiece 114 before, during, and/or after thermal processing. In some embodiments, workpiece support 112 can be transparent to and/or otherwise configured to allow at least some radiation to at least partially pass through workpiece support 112. For instance, in some embodiments, a material of workpiece support 112 can be selected to allow desired radiation to pass through workpiece support 112, such as radiation that is emitted by workpiece 114 and/or emitters 150. In some embodiments, workpiece support 112 can be or include a quartz material, such as a hydroxyl free quartz material.

Workpiece support 112 can include one or more support pins 115, such as at least three support pins, extending from workpiece support 112. In some embodiments, workpiece support 112 can be spaced from the top of the processing chamber 110. In some embodiments, the support pins 115

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and/or the workpiece support 112 can transmit heat from heat sources 140 and/or absorb heat from workpiece 114. In some embodiments, the support pins 115 can be made of quartz.

Processing apparatus 100 can include one or more heat sources 140. In some embodiments, heat sources 140 can include one or more heating lamps 141. For example, heat sources 140 including one or more heating lamps 141 can emit thermal radiation to heat workpiece 114. In some embodiments, for example, heat sources 140 can be broad-band radiation sources including arc lamps, incandescent lamps, halogen lamps, any other suitable heating lamp, or combinations thereof. In some embodiments, heat sources 140 can be monochromatic radiation sources including light-emitting diodes, laser diodes, any other suitable heating lamps, or combinations thereof. The heat source 140 can include an assembly of heating lamps 141, which are positioned, for instance, to heat different zones of the workpiece 114. The energy supplied to each heating zone can be controlled while the workpiece 114 is heated. Further, the amount and/or type of radiation applied to various zones of the workpiece 114 can also be controlled in an open-loop fashion. In this configuration, the ratios between the various heating zones can be pre-determined after manual optimization. In other embodiments, the amount and/or type of radiation applied to various zones of the workpiece 114 can be controlled in a closed-loop fashion, based on temperature of the workpiece 114.

In some embodiments, directive elements such as, for example, reflectors 800 (e.g., mirrors) can be configured to direct radiation from heat sources 140 into processing chamber 110. In certain embodiments, reflectors 800 can be configured to direct radiation from one or more heating lamps 141 towards workpiece 114 and/or workpiece support 112. For example, one or more reflectors 800 can be disposed with respect to the heat sources 140 as shown in FIGS. 2-3 and 5-6. One or more cooling channels 802 can be disposed between or within the reflectors 800. As shown by arrows 804 in FIGS. 3 and 6, ambient air can pass through the one or more cooling channels 802 to cool the one or more heat sources 140, such as the heat lamps 141.

Referring now to FIGS. 4-6, a first group of one or more heat sources 140 can be disposed on the bottom side of the processing chamber 110 and a second group of one or more heat sources 140 can be disposed on the top side of the processing chamber 110. For instance, the heat sources 140 disposed on the bottom side of the processing chamber 110 can be used to heat a back side of the workpiece 114 when it is atop the workpiece support 112. The heat sources 140 disposed on the top side of the processing chamber 110 can be used to heat a top side of the workpiece 114 when it is atop the workpiece support 112.

According to example aspects of the present disclosure, one or more dielectric windows 106, 108 can be disposed between the heat source 140 and the workpiece support 112. According to example aspects of the present disclosure, windows 106, 108 can be disposed between workpiece 114 and heat sources 140. Windows 106, 108 can be configured to selectively block at least a portion of radiation emitted by heat sources 140 from entering a portion of the processing chamber 110. For example, windows 106, 108 can include opaque regions 160 and/or transparent regions 161. As used herein, "opaque" means having a transmittance of less than 0.4 (40%) for a given wavelength, and "transparent" means having a transmittance of greater than 0.4 (40%) for a given wavelength. In some embodiments, the one or more dielectric windows 106, 108 comprise quartz, and the one or more

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opaque regions 160 comprise a higher level of hydroxyl (OH) groups than the one or more transparent regions 161.

Opaque regions 160 and/or transparent regions 161 can be positioned such that the opaque regions 160 block stray radiation at some wavelengths from the heat sources 140, and the transparent regions 161 allow, for example, emitters 150, heat sources 140, reflectance sensor 166, and/or temperature measurement devices 167, 168 to have no obstruction to radiation in processing chamber 110 at the wavelengths blocked by opaque regions 160. In this way, the windows 106, 108 can effectively shield the processing chamber 110 from radiation contamination by heat sources 140 at given wavelengths while still allowing radiation from the heat sources 140 to heat workpiece 114. Opaque regions 160 and transparent regions 161 can generally be defined as opaque and transparent, respectively, to a particular wavelength; that is, for at least radiation at the particular wavelength, the opaque regions 160 are opaque and the transparent regions 161 are transparent.

Windows 106, 108, including opaque regions 160 and/or transparent regions 161, can be formed of any suitable material and/or construction. In some embodiments, dielectric windows 106, 108 can be or include a quartz material. Furthermore, in some embodiments, opaque regions 160 can be or include hydroxyl (OH) containing quartz, such as hydroxyl (OH—) doped quartz, and transparent regions 161 can be or include hydroxyl free quartz. Hydroxyl doped quartz can exhibit desirable wavelength blocking properties in accordance with the present disclosure. For instance, hydroxyl doped quartz can block radiation having a wavelength of about 2.7 micrometers, which can correspond to a temperature measurement wavelength at which some sensors (e.g., reflectance sensor 166 and temperature measurement devices 167, 168) in the processing apparatus 100 operate, while hydroxyl free quartz can be transparent to radiation with a wavelength of about 2.7 micrometers. Thus, the hydroxyl doped quartz regions can shield the sensors (e.g., reflectance sensor 166 and temperature measurement devices 167, 168) from stray radiation of the wavelength in the processing chamber 110 (e.g., from heat sources 140), and the hydroxyl free quartz regions can be disposed at least partially within a field of view of the sensors to allow the sensors to obtain measurements at the wavelength within the thermal processing system.

A rotation system 900 is used to rotate heat sources 140. As described, the rotation system 900 is configured to rotate the heat sources 140 during processing of the workpiece 112. Thus, in certain embodiments, the rotation system 900 is used to rotate heat sources 140 while the workpiece 112 remains stationary in the processing chamber 110.

One or more exhaust ports 921 can be disposed in the processing chamber 110 that are configured to pump gas out of the processing chamber 110, such that a vacuum pressure can be maintained in the processing chamber 110. For example, process gas can flow through the processing chamber 110 according to the arrows as depicted in FIGS. 3 and 6. The process gas is exposed to the workpiece 114 and then flows around either side of the workpiece 114 and is evacuated from the processing chamber 110 via one or more exhaust ports 921. The flow of the process gas is shown by arrows 806 in FIGS. 3 and 6. One or more pumping plates 910 can be disposed around the outer perimeter of the workpiece 114 to facilitate process gas flow, which will be discussed more particularly with respect to the following figures below. Isolation door 180, when open, allows entry of the workpiece 114 to the processing chamber 110 and, when closed, allows the processing chamber 110 to be

sealed, such that a vacuum pressure can be maintained in the processing chamber 110 such that thermal processing can be performed on workpiece 114.

In embodiments, the apparatus 100 can include a controller 175. The controller 175 controls various components in processing chamber 110 to direct processing of workpiece 114. For example, controller 175 can be used to control heat sources 140. Additionally and/or alternatively, controller 175 can be used to control the heat sources 140 and/or a workpiece temperature measurement system, including, for instance, emitter 150, reflectance sensor 166, and/or temperature measurement devices 167, 168. The controller 175 can also implement one or more process parameters, such as controlling the gas flow controllers 185 and altering conditions of the processing chamber 110 in order to maintain a vacuum pressure in the processing chamber during processing of the workpiece 114. The controller 175 can include, for instance, one or more processors and one or more memory devices. The one or more memory devices can store computer-readable instructions that, when executed by the one or more processors, cause the one or more processors to perform operations, such as any of the control operations described herein.

In particular, FIGS. 1 and 4 depict certain components useful in the workpiece temperature measurement system, including one or more temperature measurement devices 167, 168. In embodiments, temperature measurement device 167 is located in a more centered location with respect to temperature measurement device 168. For example, temperature measurement device 167 can be disposed on or next to a centerline of the workpiece support 112, such that when a workpiece 114 is disposed on the workpiece support 112, temperature measurement device 167 can obtain a temperature measurement corresponding to the center of the workpiece 114. Temperature measurement device 168 can be disposed in an outer location from the centerline of the workpiece support 112, such that temperature measurement device 168 can measure the temperature of the workpiece 114 along an outer perimeter of the workpiece 114. Accordingly, the temperature measurement system includes one or more temperature measurement devices capable of measuring the temperature of the workpiece 114 at different locations on the workpiece 114. Temperature measurement devices 167, 168 can include pyrometers. Temperature measurement devices 167, 168 can also include one or more sensors capable of sensing radiation emitted from the workpiece 114 and/or capable of sensing a reflected portion of radiation that is emitted by an emitter and reflected by the workpiece, which will be discussed in more detail hereinbelow.

For instance, in some embodiments, temperature measurement devices 167, 168 can be configured to measure radiation emitted by workpiece 114 at a temperature measurement wavelength range. For example, in some embodiments, temperature measurement devices 167, 168 can be a pyrometer configured to measure radiation emitted by the workpiece at a wavelength within the temperature measurement wavelength range. The wavelength can be or include a wavelength that transparent regions 161 are transparent to and/or opaque regions 160 are opaque to, for example at 2.7 micrometers, in embodiments where the opaque regions 160 include hydroxyl doped quartz. The wavelength can additionally correspond to a wavelength of blackbody radiation emitted by workpiece 114. The temperature measurement wavelength range can include 2.7 micrometers accordingly.

In some embodiments, the temperature measurement system includes one or more emitters 150 and one or more

reflectance sensors 166. For example, in embodiments the workpiece temperature measurement system can also include an emitter 150 configured to emit radiation directed at an oblique angle to workpiece 114. In embodiments, emitter 150 can be configured to emit infrared radiation. The radiation emitted by emitter 150 may also be referred to herein as calibration radiation. In certain embodiments, the emitter 150 emits the calibration radiation onto the workpiece with a modulation in intensity, as will be further described below. Radiation emitted by emitter 150 can be reflected by workpiece 114 forming a reflected portion of radiation that is collected by reflectance sensor 166. The reflectance of workpiece 114 can be represented by the intensity of the reflected portion of radiation incident on reflectance sensor 166. For an opaque workpiece 114, the emissivity of workpiece 114 can then be calculated from reflectance of workpiece 114. At the same time, radiation emitted by the workpiece 114 can be measured by sensors in temperature measurement devices 167 and 168. In some embodiments, such radiation emitted by workpiece 114 and measured by sensors in temperature measurement devices 167 and 168 does not constitute the reflected portion of the calibration radiation that was emitted by emitter 150 and reflected by workpiece 114. Finally, the temperature of the workpiece 114 can be calculated based on radiation emitted by workpiece 114 in combination with the emissivity of workpiece 114.

Radiation emitted by an emitter (e.g., emitter 150) and/or measured by a sensor (e.g., reflectance sensor 166 and/or sensors in temperature measurement devices 167, 168) can have one or more associated wavelengths. For instance, in some embodiments, an emitter can be or include a narrow-band emitter that emits radiation such that a wavelength range of the emitted radiation is within a tolerance of a numerical value, such as within 10% of the numerical value, in which case the emitter is referred to by the numerical value. In some embodiments, this can be accomplished by a combination of a broadband emitter that emits a broadband spectrum (e.g., a Planck spectrum) and an optical filter, such as an optical notch filter, configured to pass only a narrow band within the broadband spectrum. Similarly, a sensor can be configured to measure an intensity of narrow-band radiation at (e.g., within a tolerance of) a wavelength of a numerical value. For example, in some embodiments, a sensor, such as a pyrometer, can include one or more heads configured to measure (e.g., select for measurement) a particular narrow-band wavelength.

According to example aspects of the present disclosure, one or more transparent regions 161 can be disposed at least partially in a field of view of emitter 150 and/or reflectance sensor 166. For instance, emitter 150 and reflectance sensor 166 can operate at the temperature measurement wavelength range at which the transparent regions 161 are transparent. For example, in some embodiments, emitter 150 and/or reflectance sensor 166 can operate at 2.7 micrometers. As illustrated in FIGS. 1 and 4, the transparent regions 161 can be positioned such that a radiation flow (indicated generally by dashed lines) starts from emitter 150, passes through transparent regions 161, is reflected by the workpiece 114, and is collected by reflectance sensor 166, without obstruction by window 108 (e.g., opaque regions 160). Similarly, opaque regions 160 can be disposed in regions on window 108 that are outside of the emitted and reflected radiation flow to shield workpiece 114 and especially reflectance sensor 166 from radiation in the temperature measurement wavelength range from heat sources 140. For example, in

some embodiments, transparent regions **161** can be included for sensors and/or emitters operating at 2.7 micrometer wavelengths.

In some embodiments, emitter **150** and/or reflectance sensor **166** can be phase-locked. For instance, in some embodiments, emitter **150** and/or reflectance sensor **166** can be operated according to a phase-locked regime. For instance, although opaque regions **160** can be configured to block most stray radiation from heat sources **140** at a first wavelength, in some cases stray radiation can nonetheless be perceived by reflectance sensor **166**, as discussed above. Operating the emitter **150** and/or reflectance sensor **166** according to a phase-locked regime can contribute to improved accuracy in intensity measurements despite the presence of stray radiation.

As shown in FIG. 7, an example phase locking regime is discussed with respect to plots **250**, **260**. Plot **250** depicts radiation intensity for radiation I_{IR} emitted within the temperature measurement wavelength range by emitter **150** over time (e.g., over a duration of treatment processes performed on workpiece **114**). As illustrated in plot **250**, radiation intensity emitted by emitter **150** can be modulated. For example, the emitter **150** can emit the calibration radiation onto the workpiece **114** with a modulation in intensity. For instance, the radiation intensity emitted by emitter **150** can be modulated as pulses **251**. In some embodiments, radiation can be emitted by emitter **150** in a pulsing mode. In some other embodiments, a constant radiation of emitter **150** can be blocked periodically by a rotating chopper wheel (not shown in the figure). A chopper wheel can include one or more blocking portions and/or one or more passing portions. A chopper wheel can be revolved in a field of view of emitter **150** such that a constant stream of radiation from emitter **150** is intermittently interrupted by blocking portions and passed by passing portions of the chopper wheel. Thus, a constant stream of radiation emitted by emitter **150** can be modulated into pulses **251** with a pulsing frequency corresponding to the chopper wheel rotation. The pulsing frequency can be selected to be or include a frequency having little to no overlap to operation of other components in the processing apparatus **100**. For example, in some embodiments, the pulsing frequency can be about 130 Hz. In some embodiments, a pulsing frequency of 130 Hz can be particularly advantageous as heat sources **140** can be configured to emit substantially no radiation having a frequency of 130 Hz. Additionally and/or alternatively, reflectance sensor **166** can be phase-locked based on the pulsing frequency. For instance, the processing apparatus **100** (e.g., controller **175**) can isolate a measurement (e.g., a reflectivity measurement of workpiece **114**) from reflectance sensor **166** based on calibration radiation of emitter **150** modulated at the pulsing frequency and reflected from the workpiece **114**. In this way, processing apparatus **100** can reduce interference from stray radiation in measurements from reflectance sensor **166**. In embodiments, at least one reflectance measurement can be isolated from one or more sensors based, at least in part, on the pulsing frequency.

Similarly, plot **260** depicts reflected radiation intensity IR measured by reflectance sensor **166** over time. Plot **260** illustrates that, over time (e.g., as workpiece **114** increases in temperature), stray radiation in the chamber (illustrated by stray radiation curves **261**) can increase. This can be attributable to, for example, an increasing emissivity of workpiece **114** and correspondingly a decreasing reflectivity of workpiece **114** with respect to an increased temperature of

workpiece **114**, an increased intensity of heat source **140**, and/or various other factors related to processing of workpiece **114**.

During a point in time at which emitter **150** is not emitting radiation, reflectance sensor **166** can obtain measurements corresponding to the stray radiation curves **261** (e.g., stray radiation measurements). Similarly, during a point in time at which emitter **150** is emitting radiation (e.g., pulse **251**), reflectance sensor **166** can obtain measurements corresponding to total radiation curves **262** (e.g., total radiation measurements). The reflectance measurements can then be corrected based on this information indicative of stray radiation curves **261**.

While example embodiments disclose that reflectance sensor **166** is used to collect reflected radiation that is emitted by emitter **150**, the disclosure is not so limited. In certain embodiments, one or more heating lamps **141** may be used to emit radiation similar to that of emitter **150** as described herein. For example, radiation emitted by the one or more heating lamps **141** can include a first radiation component and a second radiation component. The first radiation component emitted is configured to heat workpiece **114**, while the second radiation component emitted is modulated at a pulsing frequency. Portions of the modulated second radiation component emitted by the one or more heat lamps **141** can be reflected by the workpiece **114** and collected on the reflectance sensor **166**, such that a reflectivity measurement of workpiece **114** can be obtained.

In other certain embodiments, temperature measurement devices **167**, **168** can also be configured with sensors capable of functioning in a similar manner to reflectance sensor **166**. Namely, temperature measurement devices **167**, **168** can also collect reflected portions of a modulated radiation, such as calibration radiation, that can be used to determine a reflectivity measurement of workpiece **114**. In some embodiments, the processing apparatus (e.g., controller **175**) can isolate from reflectance sensor **166** and/or temperature measurement devices **167**, **168**, a first radiation measurement of workpiece **114** and a second reflectivity radiation measurement of workpiece **114**. The second reflectivity radiation measurement of workpiece **114** is based on a reflected portion of radiation emitted by emitter **150** or one or more heat lamps **141** modulated at the pulsing frequency.

In certain embodiments, a workpiece temperature control system can be used to control power supply to the heat sources **140** in order to adjust the temperature of the workpiece **114**. For example, in certain embodiments the workpiece temperature control system can be part of the controller **175**. In embodiments, the workpiece temperature control system can be configured to change the power supply to the heat source **140** independent to the temperature measurement obtained by the temperature measurement system. However, in other embodiments, the workpiece temperature control system can be configured to change the power supply to the heat sources **140** based, at least in part, on the one or more temperature measurements of workpiece **114**. A closed loop feedback control can be applied to adjust the power supply to the heat sources **140** such that energy from the heat sources **140** applied to the workpiece **114** will heat the workpiece to but not above a desired temperature. Thus, the temperature of the workpiece **114** may be maintained by closed loop feedback control of the heat source **140**, such as by controlling the power to the heat source **140**.

As described, the heat sources **140** are capable of emitting radiation at a heating wavelength range and the temperature measurement system is capable of obtaining a temperature measurement about a temperature measurement wavelength

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range. Accordingly, in certain embodiments the heating wavelength range is different from the temperature measurement wavelength range.

A guard ring 109 can be used to lessen edge effects of radiation from one or more edges of the workpiece 114. The guard ring 109 can be disposed around the workpiece 114. Further, in embodiments, the processing apparatus includes a pumping plate 910 disposed around the workpiece 114 and/or the guard ring 109. For example, FIG. 8 illustrate an example pumping plate 910 that can be used in embodiments provided. The pumping plate 910 includes one or more pumping channels 912, 913 for facilitating the flow of gas through the processing chamber 110. For example, the pumping plate 910 can include a continuous pumping channel 912 configured around the workpiece 114. The continuous pumping channel 912 can include an annular opening configured to allow gas to pass from a first side, such as a top side, of the workpiece 114 to a second side, such as the back side, of the workpiece 114. The continuous pumping channel 912 can be disposed concentrically around the guard ring 109. Additional pumping channels 913 can be disposed in the pumping plate 910 to facilitate gas movement within the processing chamber 110. The pumping plate 910 can be or include a quartz material. Furthermore, in some embodiments, pumping plate 910 can be or include quartz containing a significant level of hydroxyl (OH) groups, a.k.a. hydroxyl doped quartz. Hydroxyl doped quartz can exhibit desirable wavelength blocking properties in accordance with the present disclosure.

FIG. 9 depicts a flow diagram of one example method (700) according to example aspects of the present disclosure. The method (700) will be discussed with reference to the processing apparatuses 100 or 600 of FIG. 1 or 4 by way of example. The method (700) can be implemented in any suitable processing apparatus. FIG. 9 depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. In addition, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure.

At (702), the method can include placing a workpiece 114 in a processing chamber 110 of a processing apparatus 100. For instance, the method can include placing a workpiece 114 onto workpiece support 112 in the processing chamber 110 of FIG. 1. The workpiece 114 can include one or more layers comprising silicon, silicon dioxide, silicon carbide, one or more metals, one or more dielectric materials, or combinations thereof.

At (704), optionally, the method includes admitting a process gas to the processing chamber 110. For example, a process gas can be admitted to the processing chamber 110 via the gas delivery system 155 including a gas distribution channel 151. For example, the process gas can include oxygen-containing gases (e.g. O₂, O₃, N₂O, H₂O), hydrogen-containing gases (e.g., H₂, D₂), nitrogen-containing gases (e.g. N₂, NH₃, N₂O), fluorine-containing gases (e.g. CF₄, C₂F₄, CHF₃, CH₂F₂, CH₃F, SF₆, NF₃), hydrocarbon-containing gases (e.g. CH₄), or combinations thereof. In some embodiments, the process gas can be mixed with an inert gas, such as a carrier gas, such as He, Ar, Ne, Xe, or N₂. A control valve 158 can be used to control a flow rate of each feed gas line to flow a process gas into the processing

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chamber 110. A gas flow controller 185 can be used to control the flow of process gas.

At (706) the method includes controlling a vacuum pressure in the processing chamber 110. For example, one or more gases can be evacuated from the processing chamber 110 via one or more gas exhaust ports 921. Further, controller 175 can also implement one or more process parameters, altering conditions of the processing chamber 110 in order to maintain a vacuum pressure in the processing chamber 110 during processing of the workpiece 114. For example, as process gases are introduced in the processing chamber 110, controller 175 can implement instructions to remove process gases from the processing chamber 110, such that a desired vacuum pressure can be maintained in the processing chamber 110. The controller 175 can include, for instance, one or more processors and one or more memory devices. The one or more memory devices can store computer-readable instructions that when executed by the one or more processors cause the one or more processors to perform operations, such as any of the control operations described herein.

At (707) the method includes rotating one or more heat sources 140. For example, a rotation system 900 can be utilized to rotate one or more heat sources 140. The rotation system 900 can be used to rotate heat sources 140 located on either the first side of the processing chamber 110, the second side of the processing chamber 110, or both.

At (708) the method includes emitting radiation directed at one or more surfaces of the workpiece, such as a back side of the workpiece, to heat the workpiece. For example, heat sources 140 including one or more heating lamps 141 can emit thermal radiation to heat workpiece 114. In some embodiments, for example, heat sources 140 can be broadband thermal radiation sources including arc lamps, incandescent lamps, halogen lamps, any other suitable heating lamp, or combinations thereof. In some embodiments, heat sources 140 can be monochromatic radiation sources including light-emitting diodes, laser diodes, any other suitable heating lamps, or combinations thereof. In certain embodiments, directive elements, such as for example, reflectors (e.g., mirrors) can be configured to direct thermal radiation from one or more heating lamps 141 towards a workpiece 114 and/or workpiece support 112. The one or more heat sources 140 can be disposed on the bottom side of the processing chamber 110 in order to emit radiation at the back side of the workpiece 114 when it is atop the workpiece support 112.

At (710), optionally, the method includes emitting radiation directed at one or more surfaces of the workpiece 114, such as a top side of the workpiece 114, to heat the workpiece 114. For example, as shown in FIG. 4, the processing apparatus 600 can include one or more heat sources 140 disposed on the top side of the processing chamber 110 in order to emit radiation at a top side of the workpiece 114 when it is atop the workpiece support 112. The one or more heat sources 140 can include one or more heating lamps 141. Example heat sources 140 can include those previously described herein. In certain embodiments, directive elements, such as for example, reflectors (e.g., mirrors) can be configured to direct radiation from one or more heating lamps 141 towards a workpiece 114 and/or workpiece support 112.

In certain embodiments, the workpiece 114 can be rotated in the processing chamber 110 during heating of the workpiece 114. For example, the rotation shaft 900 coupled to the workpiece support 112, can be used to rotate the workpiece 114 in the processing chamber 110.

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At (712), optionally, the method includes obtaining a temperature measurement indicative of a temperature of the workpiece 114. For example, one or more temperature measurement devices 167, 168, sensors 166, and/or emitters 150 can be used to obtain a temperature measurement indicative of a temperature of the workpiece 114. For example, in embodiments the temperature measurement can be obtained by: emitting, by one or more emitters, a calibration radiation at one or more surfaces of the workpiece; measuring, by one or more sensors, a reflected portion of the calibration radiation emitted by the one or more emitters and reflected by the one or more surfaces of the workpiece; and determining, based at least in part on the reflected portion, reflectivity of the workpiece 114. In some embodiments, the workpiece reflectivity measurement can be obtained by modulating at least one of the one or more emitters at a pulsing frequency; and isolating at least one measurement from the one or more sensors based at least in part on the pulsing frequency. The emissivity of the workpiece 141 can be determined from reflectivity of the workpiece 141. In some other embodiments, one or more sensors can be used to obtain a direct radiation measurement from the workpiece 114. One or more windows can be used to block at least a portion of broadband radiation emitted by the one or more heating lamps 141 from being incident on the temperature measurement devices 167, 168 and reflectance sensor 166. The temperature of the workpiece 114 can be determined from radiation and emissivity of the workpiece 114.

At (714) process gas flow into the processing chamber is stopped and radiation emittance of heat source 140 is stopped, thus ending workpiece processing.

At (716) the method includes removing the workpiece 114 from the processing chamber 110. For instance, workpiece 114 can be removed from workpiece support 112 in processing chamber 110. The processing apparatus can then be conditioned for future processing of additional workpieces.

In embodiments, as indicated by the various arrows in FIG. 9 the method can include the listed steps in a variety of orders or combinations. For example, in certain embodiments the workpiece 114 is placed in the processing chamber 110 and exposed to radiation prior to admitting a process gas into the processing chamber 110. Further, the radiation can be emitted at the back side of the workpiece 114 and the top side of the workpiece 114 in alternating fashion, or radiation can be simultaneously emitted at the top side and the back side of the workpiece 114 in the processing chamber 110. Process gas can be admitted into the processing chamber 110 while radiation is emitted at either the top side or the back side of the workpiece 114. Further, a vacuum pressure can be maintained in the processing chamber 110 while process gas is admitted to the processing chamber 110, radiation is emitted at the top side or back side of the workpiece 114, and/or temperature measurements are obtained. Further, the heat sources 140 can be rotated by the rotation system 900 either before, during, and/or after, the steps of emitting radiation at either the front side or back side of the workpiece.

Further aspects of the invention are provided by the subject matter of the following clauses:

A method for processing a workpiece in a processing apparatus, the workpiece comprising a top side and a back side, the method comprising: placing the workpiece on a workpiece support disposed in a processing chamber; admitting one or more process gases into the processing chamber; maintaining a vacuum pressure in the processing chamber; emitting, by one or more radiative heat sources, radiation directed at one or more surfaces of a workpiece to heat at

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least a portion of a surface of the workpiece; rotating the one or more radiative heat sources; and obtaining a temperature measurement indicative of a temperature of the workpiece.

The method of any preceding clause, wherein emitting, by one or more radiative heat sources, radiation directed at one or more surfaces of a workpiece comprises emitting radiation at a top side of the workpiece.

The method of any preceding clause, wherein emitting, by one or more radiative heat sources, radiation directed at one or more surfaces of a workpiece comprises emitting radiation at a back side of the workpiece.

The method of any preceding clause, removing gas from the processing chamber using one or more exhaust ports.

The method of any preceding clause, further comprising disposing a pumping plate around the workpiece, the pumping plate providing one or more channels for the directing a flow of process gas through the processing chamber.

The method of any preceding clause, wherein the process gas comprise an oxygen-containing gas, a hydrogen-containing gas, a nitrogen-containing gas, a hydrocarbon-containing gas, a fluorine-containing gas, or combinations thereof.

The method of any preceding clause, wherein obtaining a measurement indicative of a reflectivity of the workpiece, comprises: emitting, by one or more emitters, a calibration radiation at one or more surfaces of the workpiece; measuring, by one or more sensors, a reflected portion of the calibration radiation emitted by the one or more emitters and reflected by the one or more surfaces of the workpiece; and determining, based at least in part on the reflected portion, a reflectivity of the workpiece.

The method of any preceding clause, wherein the method further comprises: modulating the calibration radiation emitted by the one or more emitters at a pulsing frequency; and isolating at least one measurement from the one or more sensors based at least in part on the pulsing frequency.

The method of any preceding clause, further comprising: blocking, by one or more windows, at least a portion of broadband radiation emitted by one or more heating lamps configured to heat the workpiece from being incident on one or more sensors.

The method of any preceding clause, further comprising stopping the flow of process gas or emitting radiation.

The method of any preceding clause, further comprising removing the workpiece from the processing chamber.

While the present subject matter has been described in detail with respect to specific example embodiments thereof, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing may readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, the scope of the present disclosure is by way of example rather than by way of limitation, and the subject disclosure does not preclude inclusion of such modifications, variations and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art.

What is claimed is:

1. A processing apparatus for processing a workpiece, the workpiece having a top side and a back side opposite from the top side, the processing apparatus comprising:

- a processing chamber having a first side and a second side opposite from the first side of the processing chamber;
- a workpiece support disposed within the processing chamber, the workpiece support configured to support the workpiece, wherein the back side of the workpiece faces the workpiece support;

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one or more pumping plates disposed around an outer perimeter of the workpiece comprising at least one continuous pumping channel comprising an annular opening configured to allow gas to pass from a top side of the workpiece to a back side of the workpiece;

a gas delivery system configured to flow one or more process gases into the processing chamber;

a first group of one or more radiative heating sources disposed on the first side of the processing chamber, the first group of one or more radiative heating sources configured to heat the workpiece from the top side of the workpiece;

a second group of one or more radiative heating sources disposed on the second side of the processing chamber, the second group of one or more radiative heating sources configured to heat the workpiece from the back side of the workpiece;

a first dielectric window disposed between the first group of one or more radiative heating sources disposed on the first side of the processing chamber and the workpiece;

a second dielectric window disposed between the second group of one or more radiative heating sources disposed on the second side of the processing chamber and the workpiece support;

a rotation system configured to rotate at least one of the first group of one or more radiative heating sources or the second group of one or more radiative heating sources; and

a workpiece temperature measurement system configured at a temperature measurement wavelength range to obtain a temperature measurement indicative of a temperature of the back side of the workpiece.

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2. The processing apparatus of claim 1, comprising one or more gas exhaust ports for removing gas from the processing chamber such that a vacuum pressure can be maintained.

3. The processing apparatus of claim 1, wherein the first dielectric window and second dielectric window comprise quartz, and the workpiece support comprises quartz.

4. The processing apparatus of claim 1, wherein the one or more radiative heating sources are configured to emit a broadband radiation to heat the workpiece.

5. The processing apparatus of claim 1, wherein the first and second dielectric windows comprise one or more transparent regions that are transparent to at least a portion of radiation within the temperature measurement wavelength range and one or more opaque regions that are opaque to the radiation within the temperature measurement wavelength range, wherein the one or more opaque regions are configured to block at least a portion of broadband radiation emitted by the radiative heating sources within the temperature measurement wavelength range.

6. The processing apparatus of claim 5, wherein the first and second dielectric windows comprise quartz, and the one or more opaque regions comprise a higher level of hydroxyl (OH) groups than the one or more transparent regions.

7. The processing apparatus of claim 1, wherein the one or more radiative heating sources are configured to emit a monochromatic radiation at a heating wavelength range, wherein the heating wavelength range is different from the temperature measurement wavelength range.

8. The processing apparatus of claim 1, wherein the workpiece temperature measurement system is configured to obtain a reflectance measurement of the workpiece.

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